Lecture Notes: Relativity

Based on Youtube series by eigenchris

Contents

1 Introduction to Relativity **2**
1.1 Gelilean Relativity 2
1.2 Special Relativity 3

2 Keys to to Relativity **7**

3 Galilean Relativity **8**
3.1 Spacetime Diagrams 8
3.2 Spacetime Interval, transforms, covariance/contravariance 9
3.3 Metric Tensor 12
3.4 Problems of Galilean Relativity 13

4 Special Relativity **13**

1 Introduction to Relativity

Before we even start by exploring relativity, either it being Gelilean or Eistein one, we should ask ourselves: what is relativity?

Relativity is how the laws of motion work in different reference frames.

1.1 Gelilean Relativity

Now, the Galilean principle of relativity, states that **the laws of motion are the same in all inertial frames of reference.**

And an inertial reference frame, is a reference frame that's not accelerating.

If we think about a very simple experiment from a scientist dropping a ball on the floor in his office room, and a car passing by this office and observing the experiment:

- The scientist will just see the ball falling down straight to the ground
- A potential car passing (at a constant velocity) by the office and looking at it, will see the ball moving along a curved trajectory towards the floor

At the end, they would disagree on the velocity, on the momentum, kinetic energy as well.
So like, what are they actually agreeing on? Well, they would agree on:

- The time passed
- The size of the ball
- The mass of the ball

Eventually, this leads to them agreeing on the laws of motion, the Newton's three laws of motion and the gravitational law:

$$\vec{F} = m\vec{a} \tag{1}$$

$$\vec{F} = G\frac{m_1 m_2}{r^2}(-\hat{r}) \tag{2}$$

And if we copy the experiment of dropping the ball inside the car passing by at constant velocity, the experiment will be exactly the same as the scientist run in his own office.
So the laws of motion are exactly the same in all inertial reference frames.

There's no experiment you can do to determine if your reference frame is stationary, or moving at constant speed. And there's no special reference frame truly stationary

And what would instead be an example of non-inertial reference frame? Well an elevator accelerating, or a car accelerating forward, a car turning a corner.
And in all these non-inertial situations, the passenger (man in the elevator, or driver in the car) always feels a misterious fictitious force pulling them in the opposite direction of the acceleration. **This means that a passenger in an accelerating reference frame, will have different set of laws of motion!** Everytime they use Newton's second law, they need to add an extra term for the fictitious force in order to predict the motion of objects correctly.

1.2 Special Relativity

This type of relativity is the one we experience when moving to speed close to the speed of light.

The story starts in the 19th centuryt with Maxwell's equations of electro-magnetism:

$$\nabla \cdot \vec{E} = \rho$$
$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$
$$\nabla \vec{B} = 0$$
$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

There's a problem with these equations, when we combine them together, we obrain equations for electromagnetic waves, or in other words..light.

$$\nabla^2 \vec{E} = \frac{1}{c^2}\left(\frac{\partial^2 \vec{E}}{\partial^2 t}\right) \qquad \nabla^2 \vec{B} = \frac{1}{c^2}\left(\frac{\partial^2 \vec{B}}{\partial^2 t}\right)$$

These equations say that light, in a vacuum, travels at the speed of $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$, and this speed, if you plug in the two constants for the megnetic permability $\mu_0 = 4\pi \times 10^{-7} H/m$, and electric permittivity $\epsilon_0 = 8.854 \times 10^{-12} F/m$, is equal to $c \approx 3 \times 10^8 m/s$.

If you think about this, and try to correlate to other speeds, like the speed of sound, you know that that's measured relative to air, or the speed of ocean's waves, that's relative to water, etc., but then what's the speed of light relative to, if it travels in the vacuum? Remember that the Galilean relativity states that there's no special reference frame! And all motion is always relative to something else.

So what are we measuring the speed of light relative to in the vacuum, if the vacuum is just empty space?? To get around this problem, scientists predicted the existence of the "aether", a special stationary reference frame that light travels in. But experiments such as the "Michelson-Morley" one showed that aether does not exist.

And this is where Einstein comes in, proposing that the speed of light c is the same in all reference frames, basically treating the speed of light as a law of physics. So in an attempt to build a new theory of relativity, Einstein began by agreeing with Galileo and said that the laws of motion are the same in all inertial reference frames, but he expanded this principle by including the Maxwell's equations in the laws of motion that the reference frames agree upon, and made the speed of light as a constant that everyone agree upon.

Finally, the two Einstein postulates of Special Relativity are:

- Laws of physics are the same in all inertial reference frames;
- Speed of light in a vacuum c is equal for all observers in inertial reference frames (a.k.a speed of light in a vacuum is included in laws of physics).

Unfortunately, these postulates are not compatible with the Galilean relativity! To see why, we can simply think about a similar experiment we thought about above, a scientist on the ground, and a car passing by at constant velocity v to the right, but this time with a giant flash light attached on its roof.

In the car's frame, the driver sees the scientist pass by with a speed of v to the left, and the light coming out from the flash light traveling to the right at speed c. But in the frame of the scientist, since the car is traveling right with speed v, the light coming out of the flash light, would appear to have the speed of light c, plus the car speed v, and this violates the postulate that the speed of light is equal to all inertial frames observers.

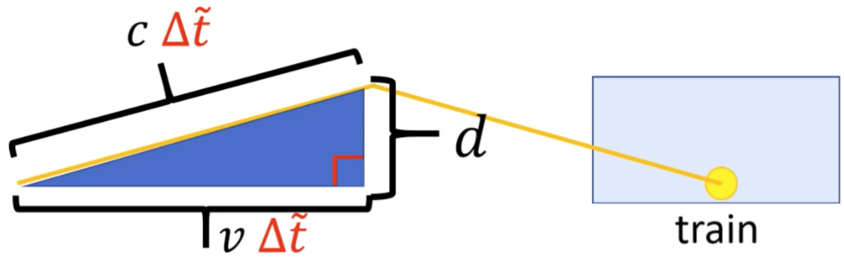
The only way to fix this, is to take time and space, which Galilean relativity stated as pointf of agreement among observers in different inertial reference frames, and change them from things that everyone agrees on, to things that everyone can disagree on.
In the theory of special relativity, time and space become relative and depend on the observer's reference frame, just like the velocity, momentum and kinetic energy.

To understand this, let's consider the example of a train passing by a station at constant speed v, this speed is measured relative to the platform on the ground where a scientist is standing. And now let's see how a beam of light behave on the train.

Imagine inside the train, a beam of light sent from one of the vagon's floors, towards the roof parpendicular to the train travel direction, and on the roof there's a mirror reflecting the light back to the floor. The distance between the floor and roof of the vagon is d.
We can use the time it takes to travel this distance as a kind of clock, so instead of ticking once per second, this clock will tick once every time tha beam crosses the train vagon in one direction. This would be given by:

$$\Delta t = \frac{d}{c} \tag{3}$$

Now consider this from the point of reference of the scientist on the platform ground. Since the train is passing by at speed v on the right, he sees the beam of light traveling in a triangle shape because of the train motion.



So in the frame of the platform, the light takes $\Delta\tilde{t}$ time to cross the train:

$$
\begin{aligned}
d^2 + \left(v\Delta\tilde{t}\right)^2 &= \left(c\Delta\tilde{t}\right)^2 \\
\frac{d^2}{c^2} + \frac{v^2}{c^2}\left(\Delta\tilde{t}\right)^2 &= \left(\Delta\tilde{t}\right)^2 \\
\frac{d^2}{c^2} &= \left(\Delta\tilde{t}\right)^2\left(1 - \frac{v^2}{c^2}\right)
\end{aligned}
$$

which means that:

$$
\begin{aligned}
\left(\Delta t\right)^2 &= \left(\Delta\tilde{t}\right)^2\left(1 - \frac{v^2}{c^2}\right) \\
\Delta\tilde{t} &= \Delta t\left(\frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}\right) = \Delta t\gamma_v
\end{aligned}
$$

This factor that you see, it's called the Lorentz factor, and we often write it as γ_v.

This tells us that, the time calculated by the scientist on the ground platform $\Delta\tilde{t}$ is actually greater than the time calculated by the moving train Δt.
In other words, the travel time for the light beam in each reference frame is not equal. When we measure the time between clock ticks for a moving clock, and measure the time between clock ticks for a stationary clock, we find that the time between clock ticks for the moving clock is larger.
This effect is called time dilation.

There is a similar effect called length contraction, where rods that are moving appear to have a shorter length than the same rods when they are standing still. However length contraction only happens along the direction of motion, in the direction perpendicular to motion, no length contraction happens.

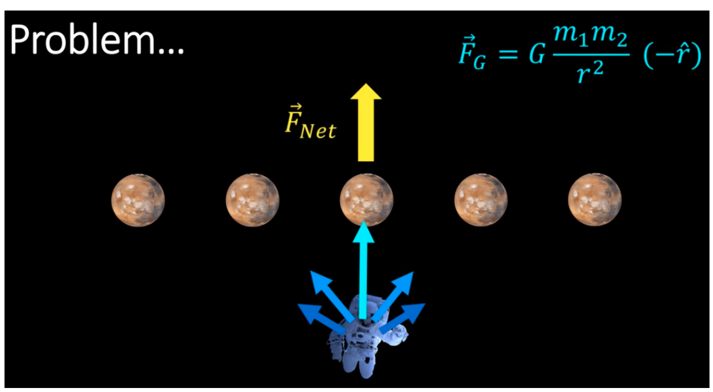
$$
\Delta\tilde{l} = \Delta l\frac{1}{\gamma_v} \tag{4}
$$

These effects are obviously only visible and noticeable when the speed is close to the speed of light, otherwise the Lorentz factor is practically one.

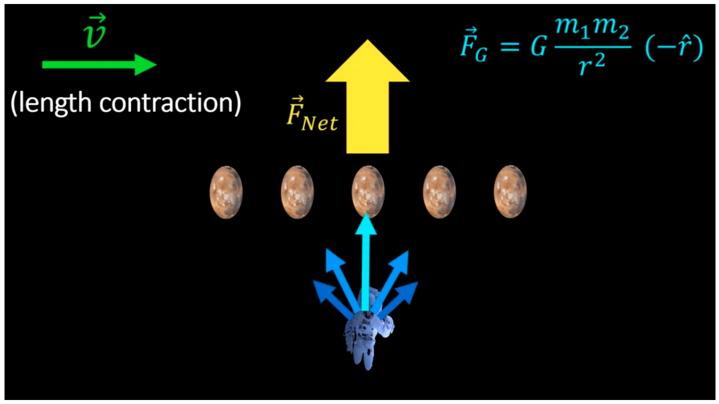
As an example, one experiment showing the time dilation is the Rossi-Hall one performed in 1941, on the muon lifetime. Muons are subatomic charged particles created in Earth's atmosphere by cosmic rays from outer space.
Liek electrons, muons are negatively charged, but unlike electrons, they're very unstable and decay on average, in $2.2\mu s$. Without considering special relativity effects, a muon would traveling at $0.995c$ would reach $656m$ on average before it decays. But this would not be enough time for a muon to reach earth from atmosphere.
At this speed, the Lorentz factor ends up being around 10, so taking now account for the time delation effect, a traveling muon will live for 10 times longer than expected lifetime from the stationary earth observer.
So this means that the muon would travel about 6.5km on average before it decays, which is indeed enough to reach the surface of our planet from the atmosphere.

If the muon is even faster, time dilation effect would be even larger, which means that the muons can travel further distances, even underground possibly. The fact that we see muons reaching Earth's surface is a real physical evidence that special relativity is correct.

Now, one last thing...
Have you ever wondered why this is called **special** relativity in the first place? Well this theory has a big problem, and that is, that's not consistent with Newtonian Gravity. If we think about a line of planets, each exerciting its force on an astronaut, according to Newton's law of gravity, the astronaut will experience a force upward by the total net force.



But if looked from a different reference frame, moving at velocity v parallel to the planets' line, these planets unergo length contraction along the line, so those become more densely packed together, and this would increase the gravitational net force on the astronaut:



The fact that the gravitational force is different in different reference frames, means that the Newton's law of gravity is not obeyed, hence the inconsistency with special relativity.

To get this right, we need a new theory of relavitiy that deals with gravity properly, which is General Relativity. End of the story is that, special relativity is valid assuming litte or no gravity, that's why it's called special. It's a special case of no gravity.

2 Keys to to Relativity

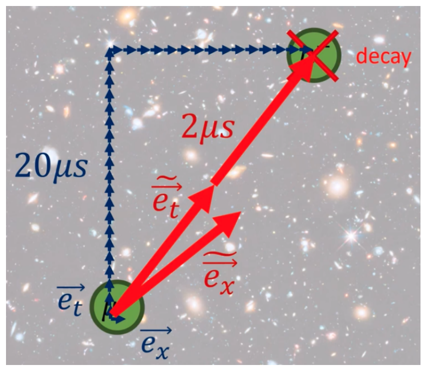
There are two main concepts we need to already touch which are very important for all relativity, from Galilean to General. These two concepts are not very hard to grasp and if you had a course in Physics or if you followed the lecture notes on Tensor for Beginners or Tensor Calculus, you already know.

- Invariance
- Covariance and contravariance

The concept of invariance is quite straight forward, and basically means that the same physical objects can be described in different ways from different points of view. Let's take a very simple example of a physical object, a pencil lying on a table. This pencil can be described using a flat 2D space using the standard orthonormal coordinate system x and y, as a vector basically. This vector can be written as a linear combination of the basis vectors e_x and e_y, so practically, its length can be measured as the sum of a scalar multiplying e_x and another scalar multiplying e_y. The scalars are also known as the components of this pencil vector, in this coordinate system.

Now, imagine we try to describe the same vector (the pencil) using a different coordinate system, which is rotated by some angle with respect to the x and y axes, on the same origin. In this new coordinate system, the vector will have different components, but the vector itself describing the pencil does not change.

This is indeed the invariance, so the same object (the pencil i.e. the vector) can be described in different ways when using different coordinate systems (basis vectors). This is also true for the muon's example of the previous section if you think about that. In its own reference frame, the muon decays in $2.2\mu s$, but in Earth's reference frame, the time it takes for his decay is 10 times larger ($\approx 20\mu s$) due to the time dilation effect.
This is not very different to explain if we use vectors and the idea of invariance. If we observe the muon creation and its later decay in space-time, the separation between these two events can be measured using vectors. With the muon coordinate system, this vector is measured to be $2.2\mu s$, but in another coordinate system, the decay time can be measured to be $20\mu s$. Time dilation in special relativity is really just the result of measuring space-time points using different basis vectors.



About the concept of covariance and contravariance, they basically mean that, when one thing grows, another thing shrinks, and the opposite changes will balance out.
This concept has been covered thoroughly on the Tensor lecture notes, but let's touch again on this briefly. When you perform any coordinate transformation, this changes both the basis and the compoents of the vectors described, and these c change in opposite ways!
Vector basis are covariant with change of coordinates, whereas vector components are contravariant. When basis changes one way, vector's components change in the opposite way.

3 Galilean Relativity

3.1 Spacetime Diagrams

Remind that Gelilean relativity states that the laws of motion are the same in all inertial reference frames.
We discussed before that, observers in different reference frames will agree on certain things, like the time and space (or space/time, Δx), the mass and indeed the laws of motion, but they will disagree on other quantities like the velocity, momentum, kinetic energy, and even the position of physical objects.

The most important quantity is actually the position of an object x, and to understand how observers disagree on positions, we'll be studying spacetime diagrams. A spacetime diagram is a tool for visualizing the motion of objects in different frames of reference.

A specetime diagram has the time t on the vertical axis, increasing towards north, and the position x as horizontal axis, increasing towards the right. Let's first try to visualize an object standing still, starting at the origin at position $x = 0$.
As time passes, if the object stays still, no movement is done on its position, so the movement is simply a straight vertical line. It's traveling through time but not through space. The path followed is called world-line.

Another example is an object moving at a constant speed of $25m/s$ to the right. If this starts at the origin, after 1 second, it will have traveled $25m$ to the right, after 2 seconds, its position will be $50m$ to the right and so on.
Its world-line is a diagonal line pointint to the upper right.

The fastest the speed, the closer the world-line is to the horizontal axis. The slower, the closer to the vertical.

Now, a really important point, remember that in Galilean relativity, all motion is relative to something else, and that there's no special reference frame that's at rest. Let's take a person, stationary, with a world-line moving only vertically through time, and a car, moving also through space at a constant speed.
From the point of view of the person, he's stationary, and the car is moving to the right. But from the point of view of the moving car, it's stationary, and the person is moving to the left, along with the road and the entire neighborhood. So, if we draw on the spacetime diagram, the person world-line as vertical, and the car moving world-line as diagonally pointing to the upper right, we're drawing the spacetime diagram from the point of view of the person.
But it would equally correct drawing the spacetime diagram from the car's point of view, where the car is stationary, and the person is diagonally moving to the upper left. Even though the two spacetime diagrams look different, they actually describe the exact same situation and the same physics, just from different points of view.

We can always re-arrange a spacetime diagram, so that any object of our choice is stationary (i.e. representing the physics from the point of view of the object). In order to do so, we have to re-arrange the time "slices" to make that object world-line vertical. There's no special stationary frame of reference, and we can always change reference frames by sliding time slices back and forth.

3.2 Spacetime Interval, transforms, covariance/contravariance

When we talk about the spacetime separation vector, or spacetime interval, we refer to the vector that measurees, in space and time, the separation between two event points. We can measure the spacetime interval using the time and space basis vectors, and different inertial frames will always agree on the time of all events, but they will disagree on the position of events, and this is because each inertial frame uses a different set of spacetime basis vectors.
Spacetime vectors are invariant, its components change based on the reference frame used to measure it.

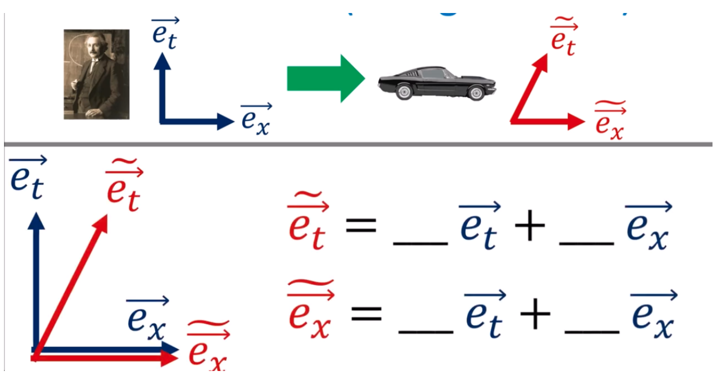
Now, we can always try to plot different events and reference frames on a spacetime

diagram to see how events are differently described when we go from one frame to another, but it would be nice to have formal equations in order to do that. In order to do that, let's try to do it with an example, considering two frames: one scientist stationary in his lab, only moving through time, and a car moving at constant speed to the right. Let's remind ourselves that, in Galilean relativity, time is universal, whereas different observers will disagree on position.
Let's say that, from the person perspective, after 1 time unit passes, the spatial separation of the car looks to be 1/2, after 2 time units, this spatial separation is 1, after 3 time units, the separation is 1.5 and so on and so forth. So we can write down:

$$\tilde{t}_{car} = t_{man}$$
$$\tilde{x}_{car} = x_{man} - 1/2 t_{man}$$

But remember that we cannot only consider the coordinate transformation, but also the basis vectors transformation, as they're both required to go a bit deeper and eventually move on to special and general relativity.

So let's try to understand how we move from the scientist's basis to the car's basis.



Just visualizing it, we can write down that:

$$\tilde{\vec{e}}_t = 1\vec{e}_t + 1/2\vec{e}_x = \vec{e}_t + 1/2\vec{e}_x$$
$$\tilde{\vec{e}}_x = 0\vec{e}_t + 1\vec{e}_x = \vec{e}_x$$

In the other direction, we can also write down:

$$\vec{e}_t = \tilde{\vec{e}}_t - 1/2\tilde{\vec{e}}_x$$
$$\vec{e}_x = \tilde{\vec{e}}_x$$

So now we have equations to move from one basis (scientist frame) to the other (car frame). In terms of math, we can also prove easily that, one trasformation is the opposite to the other.

To prove that this is also valid generically, we can substitute the 1/2 with the constant velocity the car's going v, both for the basis vectors and the components, to find out that we can write:

Basis: Man → Car

$$\begin{aligned} \tilde{\vec{e}}_t &= \vec{e}_t + v\vec{e}_x \\ \tilde{\vec{e}}_x &= \vec{e}_x \end{aligned}$$

Basis: Car → Man

$$\begin{aligned} \vec{e}_t &= \tilde{\vec{e}}_t - v\tilde{\vec{e}}_x \\ \vec{e}_x &= \tilde{\vec{e}}_x \end{aligned}$$

Components: Man → Car

$$\begin{aligned} \tilde{t} &= t \\ \tilde{x} &= -vt + x \end{aligned}$$

Components: Car → Man

$$\begin{aligned} t &= \tilde{t} \\ x &= v\tilde{t} + \tilde{x} \end{aligned}$$

The Galilean transform are basically all these formulas put together.
All of it becomes much easier if we write these down in matrix/array notation. Starting with the basis vector change from Man to Car, we can write down those in this form:

$$\begin{bmatrix} \tilde{\vec{e}}_t & \tilde{\vec{e}}_x \end{bmatrix} = \begin{bmatrix} \vec{e}_t & \vec{e}_x \end{bmatrix} \begin{bmatrix} 1 & 0 \\ v & 1 \end{bmatrix}$$

The components from Man to Car:

$$\begin{bmatrix} \tilde{t} \\ \tilde{x} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -v & 1 \end{bmatrix} \begin{bmatrix} t \\ x \end{bmatrix}$$

The two Galilean matrixes are inverse of each other, you can easily prove that by multiplying by each other and obtaining the Identity matrix.
And then, you can easily prove yourself that, the Galilean matrix used to bring the basis vectors from Man to Car is the one to use to bring the components from the Car to Man, and that the other Galilean matrix used to bring the basis vectors from Car to Man, is the one to use to bring the components from Man to Car. If you read this carefully, you will then understand the reason of calling basis vectors covariant, and components contravariant. Change-of-basis equation and Change-of-component equation use inverse Galilean matrices

because of covariance and contravariance.

When writing in matrix/array form, remember to write the covariant basis vectors as rows, and place them on the left of matrix equation, and components (contravariant) vectors as column, on the right of the matrix multiplication. This in order to have the correct formulaes coming out of them.

3.3 Metric Tensor

The metric tensor concept, in the spacetime diagrams of Galilean relativity do not make much sense, so we'll discuss briefly this concept using the 2D flat plane. Briefly because this has been covered very well in the Tensor for Beginners and Tensor Calculus lecture notes.

The metric tensor answers the question on how do we properly measure lengths in space. If you simply plot a vector on the 2D plane, you can straight forward think that, by decomponsing it as linear combination of the two basis vectors, you can apply Pythagora's theorem and call it a day.
Unfortunately this does not work consistently, and only works out the real vector length when using orthonormal basis vectors (i.e. perpendicular basis vectors with units of 1). If you try computing the length of a vector in different non-orthonormal basis, you end up with different answers.

The proper way of calculating the length is by using the dot product of the vector with itself. This works regardless of the basis vectors/reference frame utilized.

$$\|\vec{R}\|^2 = \vec{R} \cdot \vec{R} \tag{5}$$

If you write the vector as linear combination of the basis vectors you use, for example in 2D flat plane:

$$\begin{aligned}
\|\vec{R}\|^2 &= \vec{R} \cdot \vec{R} \\
&= (x\vec{e_x} + y\vec{e_y}) \cdot (x\vec{e_x} + y\vec{e_y}) \\
&= xx\,(\vec{e_x} \cdot \vec{e_x}) + xy\,(\vec{e_x} \cdot \vec{e_y}) + yx\,(\vec{e_y} \cdot \vec{e_x}) + yy\,(\vec{e_y} \cdot \vec{e_y}) \\
&= \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} \vec{e_x} \cdot \vec{e_x} & \vec{e_x} \cdot \vec{e_y} \\ \vec{e_y} \cdot \vec{e_x} & \vec{e_y} \cdot \vec{e_y} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \\
&= \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} g_{xx} & g_{xy} \\ g_{yx} & g_{yy} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}
\end{aligned}$$

where that matrix is the representation of the metric tensor, and its components are basically the dot products of the basis vectors.

Trying to tie up everything together with the concept of invariance, we can say that, the length (or length squared) of a vector $\vec{R}$ has different formulas when computed in different reference frames, but the results are always equal and invariant as long as we use the metric tensor components related to the specific basis.

The metric tensor itself is a function, which takes two vectors as input, and spits out their dot product as output. The output does not depend on the specific coordinate systems, which means that **the metric tensor itself is an invariant.** It will just have different numbers/components in different coordinate systems, but so will the vector components of the two input vectors.

Worth noticing what happens to the metric tensor in terms of covariance/contravariance, so like, whay happens to the components of the metric tensor when we shrink/grow/rotate the basis vectors from one set to another? It can be easily seen that, the metric tensor is 2-covariant, which means that when the basis vectors change, the metric tensor changes in the same way, twice.

3.4 Problems of Galilean Relativity

4 Special Relativity

References

[1] eigenchris, *Relativity*, YouTube video series,
`https://www.youtube.com/playlist?list=PLJHszsWbB6hqlw73QjgZcFh4DrkQLSCQa`